Module 1 Homework 2 Challenge

Below is a scored review of your assessment. All questions are shown.

## Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer  Partially Correct  Incorrect Answer

1

An alternative recipe for cookies requires

$$\begin{aligned} &8 \text{ eggs} \\ &6 \text{ cups flour} \\ &3 \text{ cups sugar} \end{aligned}$$

If you want to use exactly $5$ eggs to make these cookies, what will be the adjusted recipe for these cookies?

**Your Answer**

$$\begin{aligned} &5 \text{ eggs} \\ &3\frac{3}{4} \text{ cups flour} \\ &1\frac{7}{8} \text{ cups sugar} \end{aligned}$$

Correct Answer

$$\begin{aligned} &5 \text{ eggs} \\ &3\frac{3}{4} \text{ cups flour} \\ &1\frac{7}{8} \text{ cups sugar} \end{aligned}$$

## Explanation

We can clearly eliminate the first choice, because there the proportions of eggs, flour and sugar are equal. In the same way, the third choice is wrong because there the number of eggs equals the number of cups of sugar. The fourth choice is wrong because there the number of cups of flour equals the number of cups of sugar, and the last choice is wrong because the number of eggs equals the number of cups of flour. That leaves only the second choice, which is correct. If you take the amounts in the original recipe and multiply them by $\frac{5}{8}$, you will get $5$ eggs, $3\frac{3}{4}$ cups of flour, and $1\frac{7}{8}$ cups of sugar. Thinking strategically means that you don't even have to do the division here!

2 The ratio of red candies, to blue candies, to yellow candies in a bag is $2:3:5$. What percent of the total candies are red?

| | |
|---|---|
| Your Answer | $20\%$ |
| Correct Answer | $20\%$ |

## Explanation

We can use the ratios of candies, $2:3:5$, to say that the number of red candies is $2n$, the number of blue candies is $3n$, and the number of yellow candies is $5n$, where $n$ is a positive integer. The percent of the candies that is red would be the fraction

$$\frac{\#\text{ red}}{\#\text{ total}}.$$

This equals

$$\frac{2n}{2n+3n+5n}=\frac{2n}{10n}=\frac{20}{100}=20\%.$$

3 Which of the following sequences is a geometric sequence? In a geometric sequence, a term is found by multiplying the previous term by a certain number, called a factor, which is the same for all terms.

| | |
|---|---|
| Your Answer | $108,\ 36,\ 12,\ 4$ |
| Correct Answer | $108,\ 36,\ 12,\ 4$ |

## Explanation

In the first choice, the second term is $11$ times the first term, but the third term is not $11$ times the second term. We can see this because we expect the digits of $110 \times 11$ to look like $11 \times 11 = 121$, but this doesn't match our number, which has digits of $1$ only, so it cannot be a geometric sequence.

The second choice contains numbers that simply count from $1$ to $4$, which adds one each time. This isn't what we want, since the numbers aren't related to each other by a multiplicative factor. This is called an arithmetic sequence instead.

In the third choice, you can get $36$ by dividing $108$ by $3$, which is the same as multiplying it by $\frac{1}{3}$. You can get $12$ by dividing $36$ by $3$, and you can get $4$ by dividing $12$ by $3$, so this is a geometric sequence. Don't be fooled by the fact that the numbers are getting smaller! What matters is that the ratio of neighboring numbers is the same.

The fourth choice is similar to the first choice; in fact, if you divide each of the numbers in the first choice by $100$, you get the numbers in the fourth choice. This doesn't change the fact that it's not a geometric sequence, though, since the proportions should be preserved even when you scale a set of numbers by multiplying or dividing by something.

The fifth choice, like the second choice, is not a geometric sequence, though it is an arithmetic sequence, because the numbers differ by an additive factor, in this case $2$.

**4** An oak tree is $80$ feet taller than a flagpole, and three times as tall as it. How many feet tall is the flagpole? Your answer should be the number only, without units.

| | |
|---|---|
| **Your Answer** | 40 |
| Correct Answer | 40 |

## Explanation

Let the oak tree have height $o$ and the flagpole have height $f$. We are given $o = f + 80$ and $o = 3f$. Substituting the second equation into the first yields
$3f = f + 80 \implies 2f = 80 \implies f = 40.$

**5** Anna has a water bottle that is half full. After she drinks $15$ fluid ounces from the bottle, it is only one-fifth full. What is the total capacity of the water bottle, in fluid ounces?

| | |
|---|---|
| **Your Answer** | 50 |
| Correct Answer | 50 |

## Explanation

We know that $\frac{1}{2} - \frac{1}{5} = \frac{3}{10}$, so $30\%$ of the bottle is $15$ fluid ounces. This means that the full capacity of the bottle is just $\frac{15}{30\%} = \frac{15}{0.3} = \frac{150}{3} = 50$ fluid ounces.

6

Joe wants to buy a shirt that costs $\$50$, and he has a coupon for $30\%$ off. When he arrives at the store, however, he sees that the shirt has already been discounted $20\%$, but he can still use the coupon. How much does Joe end up paying for the shirt, in dollars?

| | | |
|---|---|---|
| ✓ | **Your Answer** | 28 |
| | **Correct Answer** | 28 |

## Explanation

Explanation:

Since the shirt was discounted $20\%$, when Joe arrives at the store, the shirt is being sold for $80\% \times \$50 = \$40$. When Joe uses his coupon, he gets an additional $30\%$ off, so the final price is just $70\% \times \$40 = \$28$.

**7** If two similar rectangles are proportional to each other, with the first rectangle having dimensions $\text{length}_1$ and $\text{width}_1$, and the second rectangle having dimensions $\text{length}_2$ and $\text{width}_2$, then all of the following statements are true EXCEPT for which one?

**Your Answer** $\frac{\text{length}_2}{\text{width}_1} = \frac{\text{width}_2}{\text{length}_1}$

**Correct Answer** $\frac{\text{length}_2}{\text{width}_1} = \frac{\text{width}_2}{\text{length}_1}$

## Explanation

The fractions in the first choice are both equal to the scale factor between the rectangles, in other words, the amount that you blow up or shrink the sides of one rectangle to become the other. The fractions in the fifth choice are also equal to the scale factor of the two rectangles. The fractions in the third and fourth choices correspond to a different ratio, that of the length to the width of each rectangle. For example, if you had a $2 \times 6$ rectangle, the length would be three times the width, and this would be the length-to-width ratio. Since the rectangles are similar, this ratio is the same for each, so these fractions are all the same. The second choice looks like the ratio between length and width; however, the unmatching subscripts show that you are not comparing length and width within the same rectangle. The second choice is the only one that is not true.

**8** Adam eats $20\%$ of a pizza, Bethany eats another $\frac{1}{3}$ of the pizza, and Charlie eats the remaining $14$ slices. How many slices total did the pizza originally have?

| | |
|---|---|
| **Your Answer** | 30 |
| Correct Answer | 30 |

## Explanation

The portion of the pizza that Adam and Bethany ate together was

$$\frac{1}{5} + \frac{1}{3} = \frac{8}{15},$$

and so Charlie ate

$$1 - \frac{8}{15} = \frac{7}{15}$$

of the pizza. Since he ate $14$ slices,

$$14 = \frac{7}{15} \times \text{total}$$

Solving for the total slices of pizza, we get

$$\text{total} = \frac{14}{7/15} = 15 \cdot 2 = 30.$$

**9** Which of the following is the same as $2\%$ of $50$?

| | |
|---|---|
| **Your Answer** | All of the above |
| **Correct Answer** | All of the above |

## Explanation

Let's analyze the **first choice:** Since $20\% = 2\% \times 10$, we can start with our initial expression, $2\%$ of $50$, and "move" the $10$ from the $50$ to the $2\%$ to get $20\%$ of $5$.

**Second choice:** A percent like $n\%$ is just $n$ divided by $100$, $\frac{n}{100}$. We can "transfer" the percent sign from the $5\%$ to the $20$, by doing

$$\frac{5}{100} \times 20 = \frac{5 \times 20}{100} = 5 \times \frac{20}{100} = 5 \times 20\%$$

giving us the result of the first choice, which is the same as $2\%$ of $50$.

**Third choice:** This is a direct translation of $2\%$ of $50$.

**Fourth choice:** Here we can "transfer" the percent sign from the $50\%$ to the $2$, by doing

$$\frac{50}{100} \times 2 = \frac{50 \times 2}{100} = 50 \times \frac{2}{100} = 50 \times 2\%$$

All of the answer choices are equivalent to $2\%$ of $50$, so the answer is "All of above!"

**Math note.** The analysis of the first and fourth choices uses the commutative property of multiplication!

**10** Your brother is half as tall as a tree which is three times as tall as you. What percent taller is your brother compared to you?

| | |
|---|---|
| **Your Answer** | $50\%$ |
| Correct Answer | $50\%$ |

## Explanation

Let's call your height $\text{height}$. The tree is three times your height, so the tree's height is $3 \times \text{height}$. Since your brother is half as tall as the tree, his height is $\frac{1}{2} \times 3 \times \text{height}$. Simplify to get your brother's height equaling

$$\begin{aligned} \frac{3}{2} \times \text{height} &= 1.5 \times \text{height} \\ &= (1 + 0.5) \times \text{height} \\ &= \text{height} + 0.5 \times \text{height} \end{aligned}$$

which means that your brother is taller by $0.5$ times your height, which is $50\%$.

## Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

**11** By what percent will the area of a rectangle increase if its sides increase by $20\%$? Your answer should be a number, without the $\%$ sign.

| | | |
|---|---|---|
| ✓ | **Your Answer** | 44 |
| | Correct Answer | 44 |

### Explanation

Let's get started with analyzing how our rectangle changed. Let's denote its length as $a$ and height as $b$. In our problem, $a$ and $b$ become $20\%$ bigger, so the sides of our new rectangle will be $120\%$ of $a$ and $120\%$ of $b$, which is equal to $1.2 \times \mathrm{a}$ and $1.2 \times \mathrm{b}$ accordingly. The area of the rectangle can be found by multiplying its sides. Thus the area of the new rectangle will be equal to
$1.2a \times 1.2b = (1.2 \times 1.2) \times ab = 1.44 \times ab = 1.44 \times (\text{initial rectangle area})$. This means that the new rectangle area is $144\%$ of the old rectangle area. Therefore the rectangle area increases by $44\%$.

12

You paid $\$15.73$ total for a meal at a restaurant, which includes the cost of the tip, which is $15\%$ of the menu price of the meal. What is the menu price of your meal?

| | |
|---|---|
| Your Answer | $\frac{\$15.73}{1.15}$ |
| Correct Answer | $\frac{\$15.73}{1.15}$ |

## Explanation

When you buy a meal at the restaurant, you pay for the menu price of the meal in addition to the extra $15\%$ of the menu cost, which is the tip. This means you spend $1+0.15$ times the menu cost of the meal. We can write

$$\begin{aligned} 1.15 \times \text{ meal} &= \$15.73 \\ \text{meal} &= \frac{\$15.73}{1.15} \end{aligned}$$

**13** A brother's age is $40\%$ of his sister's age. What percent of the brother's age is the sister's age? Your answer should be the number only, without units.

| | | |
|---|---|---|
| ✓ | **Your Answer** | 250 |
| | **Correct Answer** | 250 |

## Explanation

Explanation: If the sister's age is $s$ (some positive integer number), then the brother's age, $b$, will be equal to $0.4 \times \mathrm{s}$. Now we want to find what percent of the brother's age is the sister's age. Let's find out the ratio of the sister's age to the brother's age:

$$\frac{\text{sister's age}}{\text{brother's age}} = \frac{s}{b} = \frac{s}{0.4s} = \frac{1}{\frac{2}{5}} = \frac{5}{2} = 2.5.$$

This means that the sister's age is $2.5$ times the brother's age, so the sister's age is $250\%$ of the brother's age.

**14** Calculate $1001 \times 0.0142857142857142857\ldots$ as a *(terminating)* decimal.

| | |
|---|---|
| **Your Answer** | 14.3 |
| Correct Answer | 14.3 |

## Explanation

There's something special about $1001$ which we might remember even if the details escape us. It has a very neat prime factorization: $1001 = 7 \times 11 \times 13$. The number $0.142857142857142857\ldots$ is also a special repeating decimal: it is $\frac{1}{7}$. Using this, we get

$$\begin{aligned} 1001 \times 0.0142857142857142857\ldots &= 1001 \times \frac{1}{70} \\ &= \frac{11 \times 13 \times 7}{70} \\ &= \frac{11 \times 13}{10} \\ &= \frac{143}{10} \\ &= 14.3. \end{aligned}$$

**15** Number $B$ is twice as big as Number $C$, and Number $A$ is $\frac{5}{6}$ of Number $B$. Find the percent of Number $A$ that is Number $C$. Your answer should be the number only, without units.

| | | |
|---|---|---|
| ✓ | **Your Answer** | 60 |
| | Correct Answer | 60 |

## Explanation

We are given that Number $B = 2\times$ Number $C$, which means that Number $C$ = $0.5\times$ Number $B$. Also, since $A = \frac{5}{6}B$, that means that $B = 1.2 \times A$. Then we can write

$$C = 0.5 \times 1.2 \times A$$
$$C = 0.6 \times A$$
$$C = \frac{60}{100}A$$

**16** At the beginning of the year, the number of men employed at a plant was $40\%$ of the total factory workers. After six months, $6$ men were hired and $5$ women quit, making the number of men and women at the factory equal. How many people were working at the plant at the beginning of the year?

| | | |
|---|---|---|
| ✓ | **Your Answer** | 55 |
| | **Correct Answer** | 55 |

## Explanation

Let's call the initial difference between the number of men employed and the number of women employed $d$. Then, $6$ men were hired, and $5$ women quit, making the new difference equal to $0$. Let's call the number of men $m$ and the number of women $w$. Then $m + 6 = n - 5$ which $\implies 11 = 6 + 5 = n - m = d$.

Also, as we know, at the beginning of year the number of men employed at the plant was $40\%$ of the total factory workers, so the number of women employed at the plant was $100\% - 40\% = 60\%$ of the total factory workers. Thus the difference $d$ between the number of men employed and the number of women employed was $60\% - 40\% = 20\%$ of the total factory workers. This means that $11$ workers make up $20\%$ of all the workers. We remember that $20\%$ of everything is $\frac{20\%}{100\%} = \frac{1}{5}$ of this amount, so the total number of factory workers equals $5 \times (\frac{1}{5} \text{ of total amount}) = 5 \times 11 = 55$ workers.

**17** Alice and Bob are playing a game together. If Bob uses a fair dice, there is a $50\%$ chance that he will win. However, there is a $60\%$ chance that Bob will cheat using an unfair dice which gives him a $75\%$ chance of winning. What percent of the time does Bob actually win? Your answer should be the number only, without units.

| | | |
|---|---|---|
| ✓ | **Your Answer** | 65 |
| | Correct Answer | 65 |

## Explanation

Explanation:

Bob can win while cheating, or he can win fair and square. If he wins while cheating, the probability is $\frac{60}{100} \times \frac{75}{100}$, and if he wins fair and square, the probability is $\frac{40}{100} \times \frac{50}{100}$, so adding these gives

$$\frac{4 \times 5 \times 10 \times 10}{100 \times 100} + \frac{6 \times 75 \times 10}{100 \times 100}$$

$$20 + 45 = 65.$$

## Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

**18** A watermelon is $99\%$ water. You have $100$ pounds of watermelon. After a week under the sun, the watermelon has dried down to being $98\%$ water. What is the total weight of the watermelon now? Your answer should be the number only, without units.

| | |
|---|---|
| **Your Answer** | 50 |
| Correct Answer | 50 |

### Explanation

The main idea of this problem is to understand that after a week under the sun, the watermelon has **a new** weight, $98\%$ of which is **water,** so the new weight is not $100$ pounds.

As we know, the amount of water and, as a result, the amount of watermelon, has changed after a week under the sun, so to solve our problem let's find something that was unaltered throughout all this. Call the part of the watermelon that does not contain a water at all "dry," so this "dry" part remains the same weight after a week in the sun. At the beginning, the "dry" part of the watermelon was $100\% - 99\% = 1\%$, so the entire watermelon is $100 \times (\text{dry})$, and it was $100$ pounds, so the "dry" part was $\frac{100 \text{ pounds}}{100} = 1 \text{ pound}$. Then, after a week, the "dry" part of the watermelon became $100\% - 98\% = 2\%$, so $1$ pound became $2\%$ of the new watermelon, which is $\frac{1}{50^{\text{th}}}$; thus, the weight of the new watermelon is $1 \text{ pound} \times 50 = 50 \text{ pounds}$.

**19**

Evan is deciding whether to open a bank account in Bank $A$ or Bank $B$. Bank $A$ has a yearly annual interest rate of $10\%$ (that is, at the end of every year, the amount of money in a Bank $A$ account increases by $10\%$. Bank $B$ has an interest rate that is applied once every three years (that is, at the end of every three-year period, the amount of money in a Bank $B$ account increases by some fixed rate). What is the smallest such interest rate that Bank $B$ must have in order to be competitive with Bank $A$?

Express your answer as a percent rounded to the nearest tenth, e.g. 50.5

| | | |
|---|---|---|
| ✓ | **Your Answer** | 33.1 |
| | **Correct Answer** | 33.1 |

## Explanation

After three years, the amount of money in a Bank $A$ account will increase by $(1+0.1)^3 = 1.1^3 = 1.331$. This means that every three years, there is essentially an interest rate of $33.1$ in a Bank $A$ account, so Bank $B$ should set their interest rate to be at least as high as this number.

**20** Near a lake, $17\%$ of the pollywogs eat $75\%$ of the flies. If they had each eaten $7\%$ more, what percent less do the rest of the pollywogs eat?

| | |
|---|---|
| **Your Answer** | 28 |
| Correct Answer | 21 |

## Explanation

Explanation:

**Solution 1.** Calculate consumption as flies per pollywog. For the $17\%$ of pollywogs, they eat $75\% \times 1.07$ percent of the flies, or $80.25\%$ of the flies. This leaves $19.75\%$ of the flies left for the remaining pollywogs, who before enjoyed $25\%$ of the flies, so this is a reduction of $25\% - 19.75\% = 5.25\%$ in their fly consumption. To get the percent less that the other pollywogs eat, we want the decrease divided by their previous consumption, or $\frac{5.25}{25} = 0.21$, so the answer is $21\%$.

**Solution 2.** Notice that the first solution didn't use the $17\%$ in the calculation. Sometimes looking at the relative sizes of the numbers can simplify things. The well-fed pollywogs ate three times as many flies as the not-well-fed pollywogs, $75\%$ to $25\%$. If they increase their consumption by $7\%$, then the effect on decreasing the not-well-fed pollywogs' consumption is by a factor of three, so they get $7\% \times 3 = 21\%$ fewer flies.

Summary

Software by
brillium
Version 11.2

Privacy Policy. Assessment content is copyright 2026, Expii, Inc..